

On Portable Irrigation Systems

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Abstract

We model the water distribution for a single sprinkler and integrate it with a custom pressure / volume flow rate calculator to determine water distributions for arbitrary pipe set configurations. We use the information to determine an optimal tradeoff between the Christiansen Uniformity index and operation time.

The single-sprinkler water droplet distribution is modelled using ballistics theory combined with catch-can metrics. Droplet distributions from a single nozzle are modelled using empirical data. The subsequent model is then improved by integrating drag forces, wind drift and evaporative losses into the ballistics system. A high-resolution computer model is developed using numerical integration techniques, namely a Runge-Kutta fourth-order approximation.

The pressure / volume flow calculator was developed assuming laminar flow and Newtonian fluid dynamical models in order to integrate pipe interactions. The model was calibrated against empirical data.

The pipe-set tester was developed under a linear superposition model. Three pipe-set arrangements, across the field, were tested to determine the optimal system with a uniformity greater than 80% and maximal time efficiency.

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1 Introduction and Assumptions

Efficient irrigation of an arbitrary area of farmland is a difficult problem with many proposed historical solutions. Uniformity of water distribution and minimal time requirements are unquestionable required parameters [1]. In this paper we analyze a hand-move irrigation system, which is often used on small ranches with aims to:

- optimize maintenance time (time that rancher has to spend to maintain the irrigation system).
- optimizes uniformity of irrigation (indexed by Christiansen's uniformity (CU)) [2].

The main problem of irrigation is the distribution of water from a central pipe over a large surface area. In the given framework a source, also known as the hydrant, supplies water at a pressure of 420 kPa and volumetric flow rate of 150 liters per minute. The source is connected to the source line pipes which give access to water in the field. The "pipe-set" of 20 meter length is connected to source line and has sprinklers which irrigate the field. Both the pipe set and the source line has diameter 10 cm. The sprinklers receive a given flux of liquid, which they spread over an area. The rancher relocates pipe-sets according to an optimal schedule, which this research attempts to find by varying the following parameters:

- the position of the source line
- the position of the pipes
- the number and distribution of the sprinklers
- schedule of irrigation.

The following uncontrollable parameters will be included in the analysis of the irrigation system:

- the decrease of pressure due to friction in the source line and pipe-set.
- evaporation of the water droplets.
- wind effects on rate of evaporation

In the analysis we also assume the following conditions:

- water is incompressible
- water is a newtonian fluid
- the water flow is laminar.
- the source line and pipes are in perfect conditions
- average temperature of 25°C
- wind does not affect the trajectory of the particle.

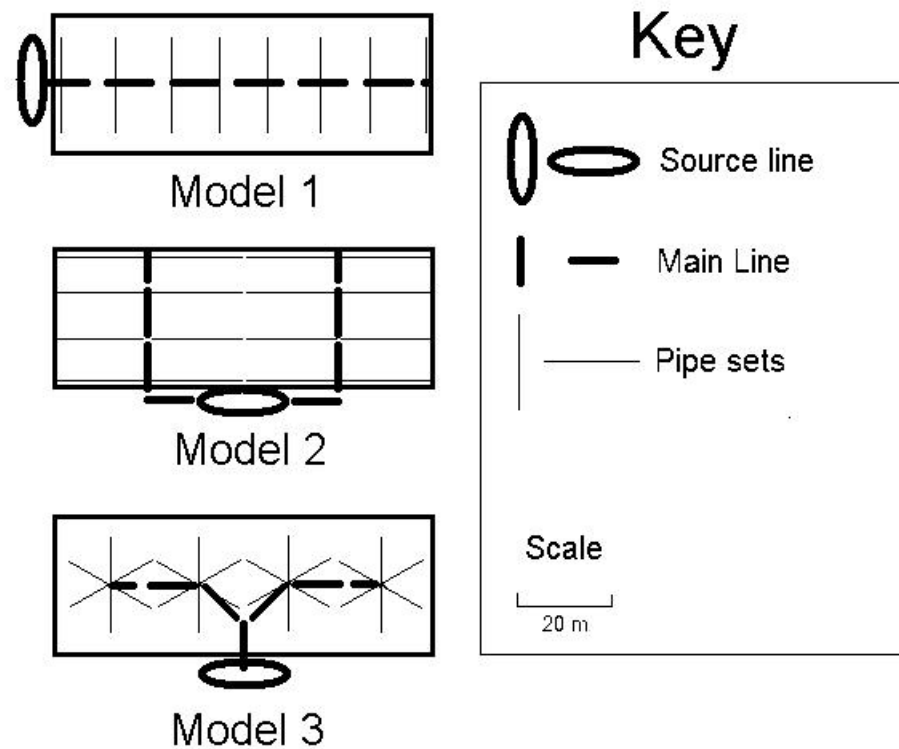
The first assumption is necessary to invoke conservation of volume flow rate (for a compressible fluid can compress, resulting in times at which the inlet volume flow rate exceeds the outlets' volume flow rates) The second assumption is true by default (as some sources define Newtonian fluid in terms of water), allowing us to write $\tau = \mu \frac{dv}{dx}$, where $\frac{dv}{dx}$ is the change in velocity along the path and μ is the viscosity.

The third assumption can be easily justified by the fact that because aluminum has low resistance and water is an incompressible fluid, in particular a newtonian fluid. The last three assumptions are made to make a model tractable; no single model will be able to consider all the possibilities (as evidenced by the intractability of the Navier-Stokes equations). Note that a strong gust of wind affects the trajectory of the particle, but for winds less than $2 \frac{m}{s}$ wind has negligible effect [2].

In section 2 we discuss various configurations of the source line, the pipe-set and position of sprinklers. Section 3 discusses the programs which were written to simulate the irrigation process. Section 4 will discuss the scheduling.

2 Structure of the System and Scheduling

Three models were chosen for a ground pipe system, as shown below:



The two main considerations are water distribution uniformity and pipe length. A longer pipe is not completely uniform, as discussed in the next section. The ground pipe in model 1 extends 80 meters into the field. The ground pipe in model 2 extends 60 meters into the field. There is a smaller ground pipe in model 3 but experimentation revealed unequal water distributions. Also, rotation of the pipe-sets is less time-efficient than linear movement.

There are two ways we could optimize model 1 in terms of the configurations of pipe-sets and the schedules. The first example shows the result for 10 sprinklers uniformly distributed across the pipe-set (every two meters), placed every six meters in each 30'x20' sector of the field. The second example shows the result for 4 sprinklers uniformly distributed on a pipe, also

with 5 stops per quadrant. These are the best schedules according to our model, when optimizing the Christiansen Uniformity. The first model has uniformity 87% and the second model has uniformity 84%.

The optimal sprinkler orientation was discovered through experimentation with varying sprinkler and pipe-set distributions. Through each tour of the entire field, at most .75 cm of water can fall in a given area; thus, in order to add 2 cm water, there must be at least 3 passes through the field. This model left parts of the field with fewer than two centimeters of water; In order to compensate, we require six passes through the field with the aforementioned pipe-set configuration and placement.

3 Simulation

Simulation code was written using Dev-C++, an IDE for the GNU C++ MinGW compiler on Windows. The modelling was broken down into several steps, and the results were compiled into the large model. We wrote an application to calculate the distribution for a single sprinkler, the pressure gradient along a pipe system, and the total distribution among multiple sprinklers connected in tandem to the pipe system.

3.1 Single-Sprinkler System

In order to fully understand the effects of many sprinklers, we needed to develop a model to understand one sprinkler in isolation. Ignoring wind and evaporative losses, we can start from a simple ballistic model. We can break down the simple model into several steps, each step building upon the previous results to produce a more accurate model:

- (a) The simple ballistic model, in which the only force is gravity, is a natural starting point. In this model, the droplets were like little balls, subject to the force $F = mg$. In 3 dimensions (where z is in the direction of the azimuth) we have 3 equations: $F_x = F_y = 0$ and $F_z = m\ddot{z} = mg$. This is an elementary physics problem, with a well-known analytical solution. Because the first-order droplet trajectories are not affected by the superposition of distributions [3], we are justified in considering a single sprinkler and summing results across all sprinklers.

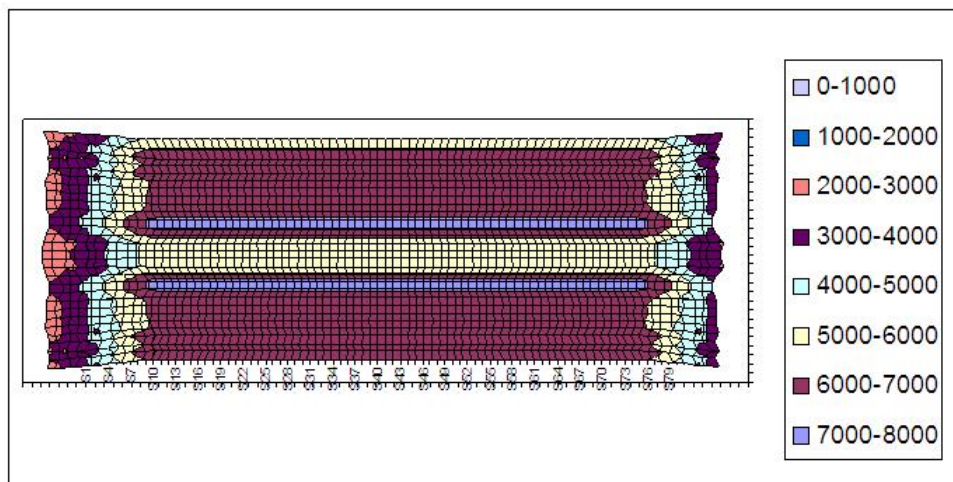


Figure 1: Each box is $1m^2$. The units of this picture are mL

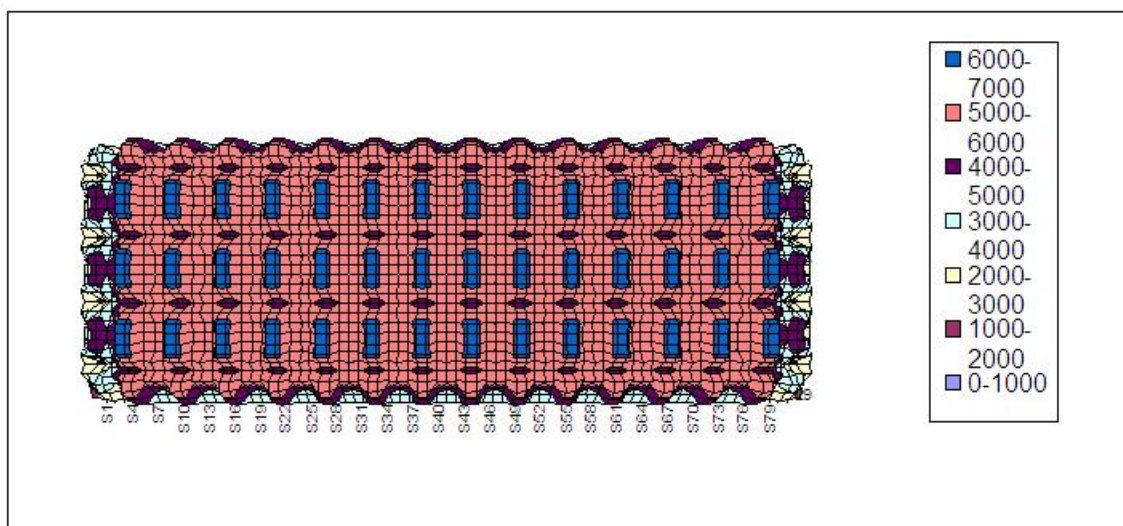


Figure 2: Each box is $1m^2$. The units of this picture are mL

- (b) The drag modification introduces F_r , the drag force. This force is in the opposite direction of the current velocity, with a magnitude inversely proportional to the square of the resistance. In particular, $F_r = \frac{1}{8}\rho_a C \pi D^2 |\vec{V}|^2 = m C_2 V^2$, where m is the mass, V is the velocity in air, ρ_a is the density of water, D is the diameter, and C is the drag coefficient [3]. For the water flowing from aluminum pipes into air, the drag coefficient C is estimated at .44 [3]. This is an excellent second-step, because analytical solutions are still possible (the differential equation $\ddot{x} = k\dot{x}^2 + c$ has a solution through partial fraction decomposition). The differential equations and solutions are:

$$\begin{aligned} m\ddot{\mathbf{x}} &= -k\dot{\mathbf{x}}^2 \\ m\ddot{\mathbf{y}} &= -k\dot{\mathbf{y}}^2 - ng \\ x(t) &= \frac{m}{k} \ln\left(\frac{v_{0,x}k}{m}t + 1\right) \\ y(t) &= h - \frac{m}{k} \ln\left(\frac{1/\sqrt{1 + \frac{k}{gn}v_{0,y}}}{\cos(\arctan(\sqrt{\frac{k}{gn}v_{0,y}}) - t\frac{\sqrt{kng}}{m})}\right) \end{aligned}$$

where $\mathbf{x}, \mathbf{y}$ are the displacements in the x and y dimension, respectively. The initial conditions here are $x(0) = 0, \dot{x}(0) = v_{0,x}, y(0) = h$, and $\dot{y}(0) = v_{0,y}$.

- (c) Whereas the first two cases allowed for straight independent analytical solutions, integrating wind into the equations leads to dependencies among the terms. This is due to the equations of motion when wind is involved. Supposing that there is no wind component normal to the ground, we get the following acceleration equations:

$$\begin{aligned} \ddot{x} &= -\frac{3}{4} \frac{\rho_{air}}{\rho_{H_2O}} \frac{C}{d} w(\dot{x} - w_x) \\ \ddot{y} &= -\frac{3}{4} \frac{\rho_{air}}{\rho_{H_2O}} \frac{C}{d} w(\dot{y} - w_y) \\ \ddot{z} &= -\frac{3}{4} \frac{\rho_{air}}{\rho_{H_2O}} \frac{C}{d} w\dot{z} - g \end{aligned}$$

where ρ is density, d is the drop diameter, w_x is the wind component in the x direction, and $w = \sqrt{w_x^2 + w_y^2}$ is the total wind. This is not

analytically tractable without intensive computation, so a numerical integration approach was used (as discussed below). As for losses due to wind drift and evaporation, the formula $20.44 + 2.7W$, where W is wind-speed in metres per second, was empirically found in [1].

3.2 The Pressurizer

In order to quantify the effects of the pipe on the pressure drop, a pressurizer simulator was created (this is used in tandem with the single-sprinkler simulator to produce the net effect). The pressure calculations are relatively simple:

Because water is incompressible and we are considering laminar flow, $\frac{\partial P}{\partial L}$ is constant (that is to say, the pressure drop is linear with respect to the length of the pipe traversed). Because we are dealing with a Newtonian fluid, we have $\tau = \mu \frac{dv}{dx}$, so $d^2 \frac{\pi}{4} \delta p = DL\tau\pi$ and, after some simplifications, we see that $\delta p = C \frac{L}{d} \frac{\rho_{water}}{2} \bar{v}^2 = \frac{8L\rho_{water}Q^2}{d^5}$, where Q is the volume flow rate.

The drop in pressure at the interface between pipes can be calculated by appending an ersatz pipe of length $\frac{Kd}{4C}$, where $K = .62$ for a contraction of cylindrical pipes.

Equations at hand, we developed a program to accept an initial flow rate and pressure as well as a series of pipes (each of which have a length and a diameter). The program returns the pressure at each of the points. The following table summarizes the extent to which pressure dropped:

n	$P^\circ(Pa)$	$L(m)$	$d(m)$	$P'(Pa)$
1	420000	20	0.1	415542
2	415542	30	0.1	408855
3	408855	20	0.1	404397

Table 1: Pressure drop for a nine-element pipe-set with varying diameters, calculated using Bernoulli's principle and formulae.

3.3 Mass and Diameter Variance

[4] gave a deterministic function for determining the position of the landing point of the particle given the actual mass and the diameter. We approximated the landing point in the same way that experimentalists measure water

propagation: catch cans. Motivated by the literature, the following models for variance were chosen (the sprinkler rotated 50 times with 800 shots per rotation – at $n\frac{\pi}{400}\forall n \in \mathbb{Z}_{800}$ – with a fixed firing frequency of 20 hz):

- Each drop has equal mass (calculated through volume flow rate), radius with a uniform distribution: This distribution has resulted in a massive dry patch in the middle. Along one cline, we found the number of water drops (out of 80000) to be $\langle 0, 0, 0, 0, 0, 0, 0, 0, 0, 20, 168, 145, 119, 114, 119, 125, 112 \rangle$. This is clearly not a great distribution, as scaling shows that there is a ring of intense rain and light sprinkling outside the inner angle
- Each drop has equal mass, radius with normal distribution: This distribution has not resolved the patch situation. As it turns out, no fixed mass solution can resolve the issue.
- Each drop has equal mass, radius with log-normal distribution: This distribution was motivated by the exponential form of the radius distribution, as advocated by [5]. This leads to a more uniform distribution, with a CV of 80%, but it still falls plague to the hole issue. This will be resolved from a neat trick in the literature:
- Mass and Radius are seen as functions of pressure, data calibrated with volume flow rate. The main formula is:

$$m = (r * 1000 / 1.14)^{1/.361}$$

with a distribution of the form $D_i \equiv a_i P^{b_i}$ (justifying the relative accuracies of the two forms). It is this final form that is used.

3.4 Numerical Integration Strategies

In approximating the differential system for the single particle with wind and drag, a numerical integrator was needed. There were four styles tried:

- Newton's method - gave weak results when compared to actual values for simple cases. This involved a first-order approximation.
- Euler's method - second-order approximation, more accurate but not as efficient (and susceptible to drastic changes with a simple reordering of operations, as calculating average velocity after updating velocities gives different answers from calculating average velocity before updating velocities).

- Modified Romberg integration - there is no nicer form to massage the expression (that is to say, the accuracies gained in calculating the intermediate values are drastically exceeded by the time required to run the command and by the accuracies of a comparably-timed Runge-Kutta fourth-order approximation).
- Runge-Kutta fourth-order approximation - This method uses carefully chosen intermediate values, based on the function itself, to give a nice approximation. Appropriately weighted (in this case, since we know there is an log-normal distribution, we use a logarithmic weighting scheme), the accuracy is fourth-order in the step size. Furthermore, unlike Euler's method, there is very little change when swapping the order of operations ("average then change" as opposed to "change then average").

These results are not surprising: The SIRIUS system also reported excellent results using Runge-Kutta fourth-order approximation.

4 Conclusions

The Christiansen uniformity for this model is greater than 83%. This method requires 20 steps per field pass (there are 4 20'x30' sectors, each of which require 5 steps). In order to guarantee the minimum required amount of water, the farmer must make at least three passes, for a total of 60 stops. Each stop lasts about half an hour (to reach .75 cm height, there must be at least $7.5mm \times m^2 = 7.5L.$), for a total of 30 hours over the course of 4 days (less than 8 hours a day).

To save some additional time during the day, the farmer could set the hand-move system up at night, using an electronic timer to control water release.

References

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